

Docker Container Logs Missing SOP

Objective

Restore Docker log visibility when container logs are not available through docker logs or monitoring tools due to logging driver, stdout/stderr, or collector issues.

Scope

Applicable to Docker-based services, containerized application deployments, DevOps teams, SRE teams, platform teams, and synthetic incident workflows across development, staging, and production-like environments.

Pre-requisites

- 1. Access to Docker host, Docker CLI, container runtime logs, and the target container or image name.
- 2. Permission to inspect containers, images, volumes, networks, resource limits, and compose files where applicable.
- 3. Access to application logs, deployment artifacts, Dockerfile, environment variables, and registry credentials.
- 4. Monitoring access for host CPU, memory, disk, container restart count, and container health checks.
- 5. Escalation contact for application, DevOps, security, registry, infrastructure, and platform teams.

Incident Metadata

Field	Value
Ticket ID	T044
Issue	Docker container logs missing
Category	Monitoring
Service	Docker
Severity	Low
Status	Resolved
Description	Logs not visible
Expected Resolution	Check logging driver configuration

Related Log Evidence

Log ID	Timestamp	Level	Message	Error Code
N/A	N/A	N/A	No direct log evidence found in logs CSV for this ticket.	N/A

Procedure

- 1. Confirm affected container, logging command, expected log source, and time window.
- 2. Check whether the application writes logs to stdout/stderr or only to internal files.
- 3. Inspect Docker logging driver configuration for the container and Docker daemon.
- 4. Check log rotation, disk usage, and whether log files were removed or truncated.
- 5. Validate log collector agent status if logs are shipped to monitoring tools.
- 6. Update application logging to stdout/stderr or configure file log collection.
- 7. Correct logging driver configuration and recreate container if required.
- 8. Generate a test log and confirm it appears in docker logs and monitoring dashboards.

Common Commands

```
docker ps -a
docker logs <container-name-or-id> --tail=100
docker inspect <container-name-or-id>
docker stats --no-stream
docker images
docker network ls && docker network inspect <network-name>
docker volume ls && docker volume inspect <volume-name>
docker system df
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
docker logs empty	App logs only to file	Write to stdout/stderr or collect file logs
Logs truncated	Log rotation or disk cleanup	Adjust rotation and retention
Logs missing in dashboard	Collector agent issue	Restart or fix log collector
docker logs unsupported	Non-json logging driver	Check logging driver configuration

Best Practices

1. Use stdout/stderr for containerized application logs.
2. Configure log rotation to prevent disk pressure.
3. Monitor logging agent health.
4. Document logging driver choices per environment.

References

Docker Docs | Internal Tickets CSV | Internal Logs CSV | Container Runtime Logs | Monitoring Dashboards | DevOps Runbook